```
Python 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/krish/AppData/Local/Programs/Python/Python313/Dry run 1st
que.py

Enter Start Node : A
Enter Goal Node : C

======================================
Iteration : 1
Current Node : A
Open List : []
Closed List : []
g(n) = 0
h(n) = 7
f(n) = 7

======================================
Iteration : 2
Current Node : B
Open List : [('C', 8)]
Closed List : ['A']
g(n) = 2
h(n) = 6
f(n) = 8

======================================
Iteration : 3
Current Node : E
Open List : [('C', 8), ('C', 9), ('D', 12)]
Closed List : ['A', 'B']
g(n) = 4
h(n) = 2
f(n) = 6

======================================
Iteration : 4
Current Node : C
Open List : [('C', 9), ('D', 9), ('D', 12), ('C', 11)]
Closed List : ['A', 'B', 'E']
g(n) = 4
h(n) = 4
f(n) = 8

Goal Reached
Optimal Path : A -> C
Total Cost : 4
```

```
Search Again?(y/n): y

Enter Start Node : B
Enter Goal Node : D

======================================
Iteration : 1
Current Node : B
Open List : []
Closed List : []
g(n) = 0
h(n) = 6
f(n) = 6

======================================
Iteration : 2
Current Node : E
Open List : [('C', 7), ('A', 9), ('D', 10)]
Closed List : ['B']
g(n) = 2
h(n) = 2
f(n) = 4

======================================
Iteration : 3
Current Node : C
Open List : [('D', 7), ('A', 9), ('D', 10), ('C', 9)]
Closed List : ['B', 'E']
g(n) = 3
h(n) = 4
f(n) = 7

======================================
Iteration : 4
Current Node : D
Open List : [('A', 9), ('C', 9), ('D', 10), ('A', 14)]
Closed List : ['B', 'E', 'C']
g(n) = 4
h(n) = 3
f(n) = 7

Goal Reached
Optimal Path : B -> E -> D
Total Cost : 4

Search Again?(y/n): n
```